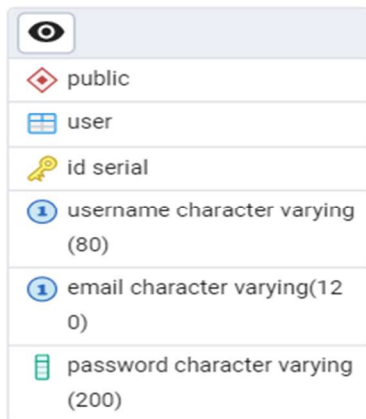


Basic ER diagram:



Details of Hardware & Software :

Hardware Requirements:

- 4GB RAM minimum
- 500MB disk space
- Microphone for STT testing (optional)

Software Requirements:

- Python 3.9+
- Node.js 18+
- Vite + React + TailwindCSS
- Flask + Flask-JWT-Extended
- Dependencies: PyPDF2, gTTS, SpeechRecognition, PyMuPDF, FPDF

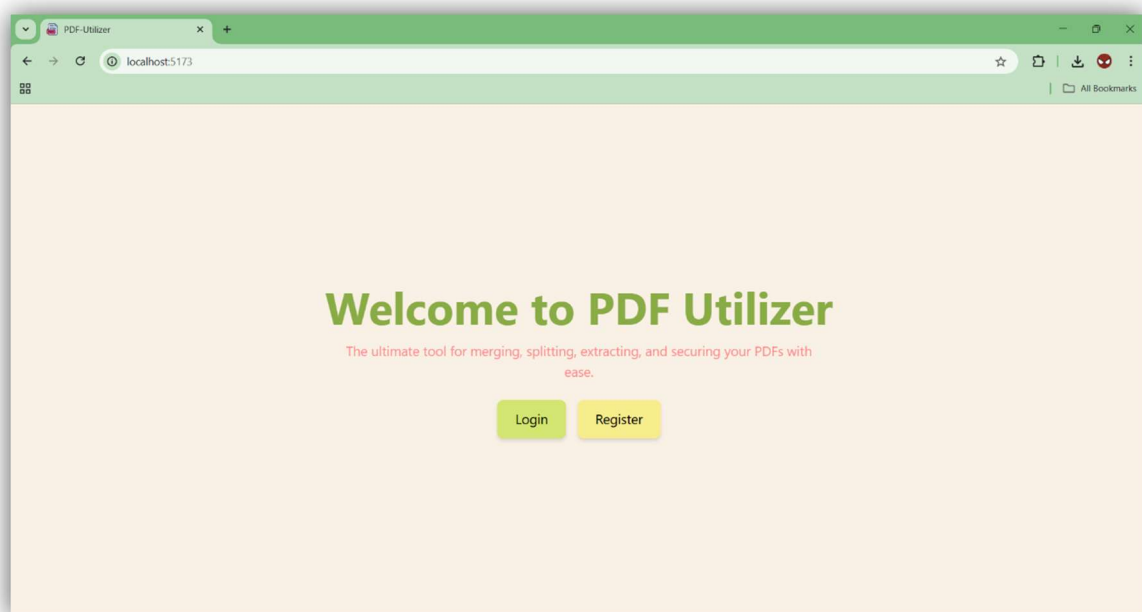
3.4 Experiment and Results for Validation and Verification

The platform was tested with various file types and scenarios:

- Large file merging: Successfully merged 50MB+ PDFs.
- Audio file recognition: STT correctly identified spoken content from `.wav` and `.webm` formats.
- Language translation: Maintained text integrity while translating paragraphs across languages.
- User authentication: Token-based login prevented unauthorized access to tool pages.

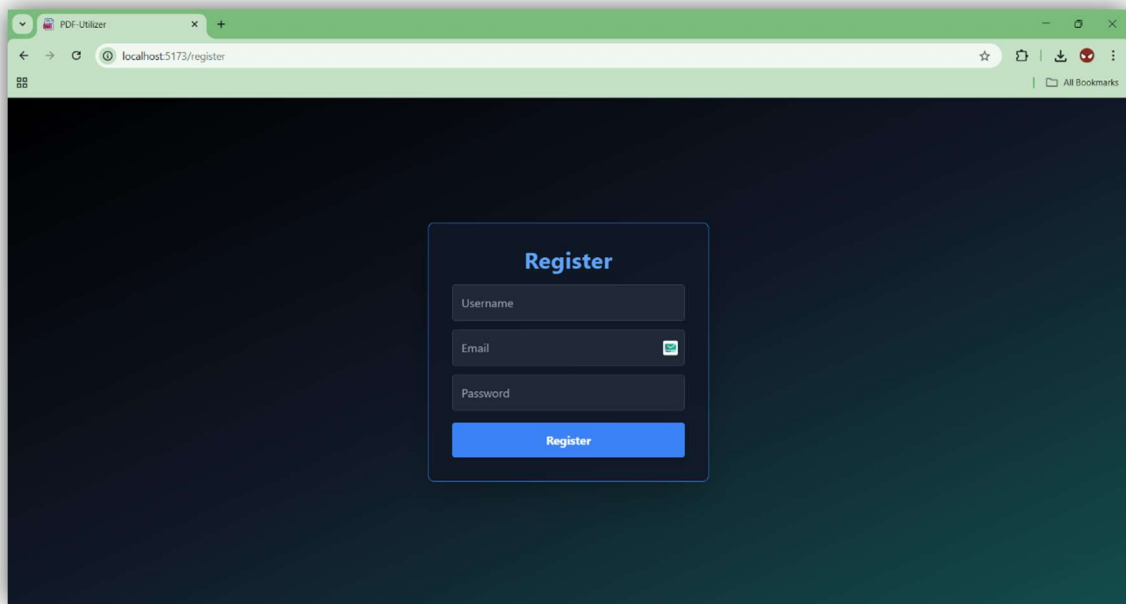
Screenshots:

1.



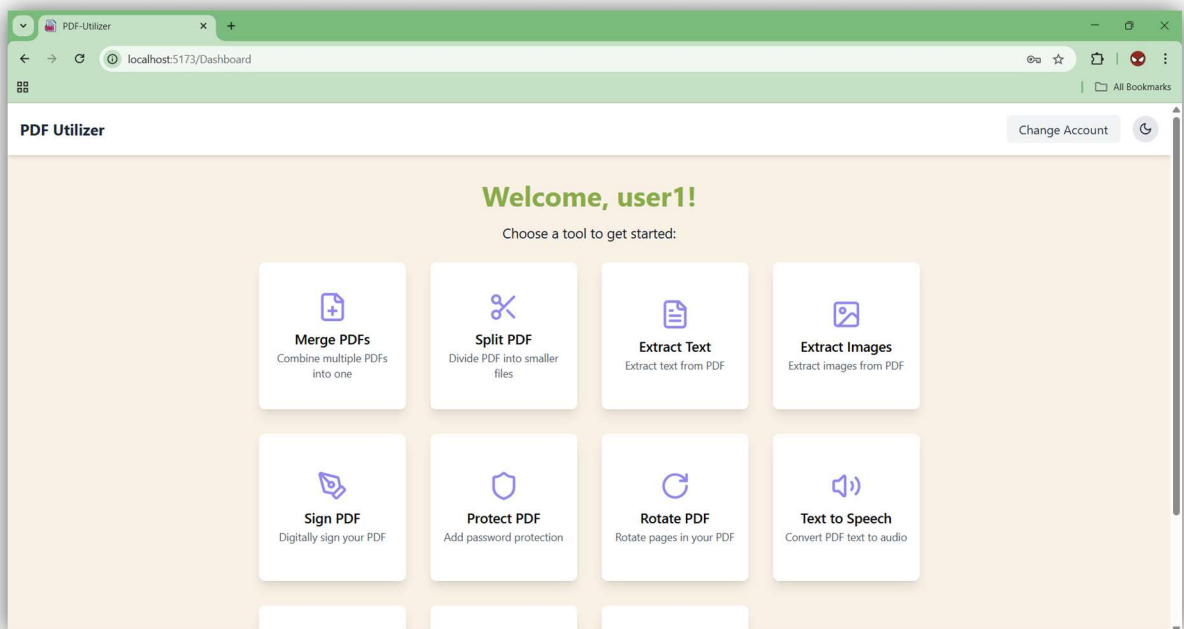
This is the landing page a one-stop platform offering powerful and free PDF tools with a user-friendly interface for easy access and navigation.

2.



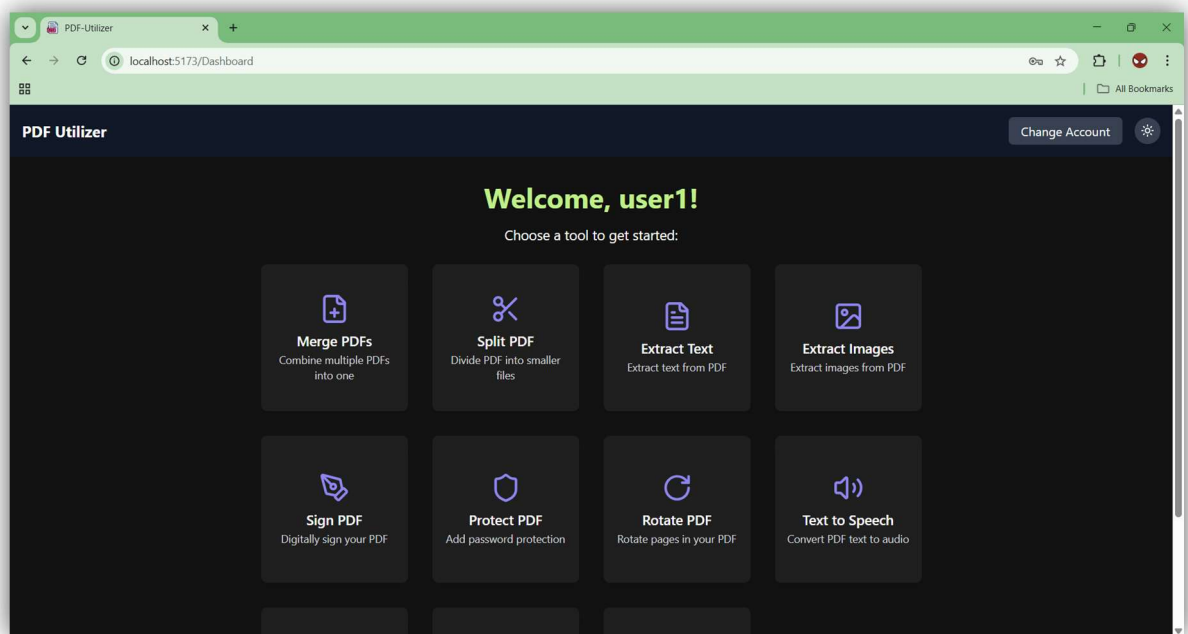
This is the register page of PDF-Utilizer, where new users can securely create an account to access all the PDF tools and features provided by the platform.

3.



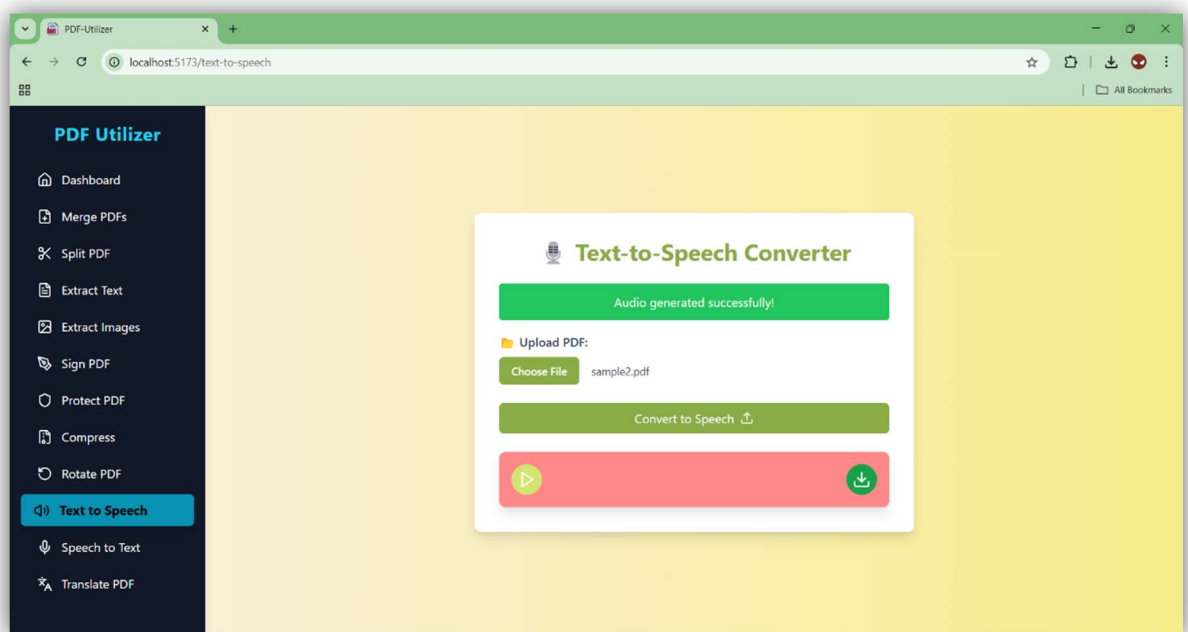
This is the Dashboard of PDF-Utilizer, serving as the central hub where users can access all available PDF tools through an intuitive and organized interface.

4.



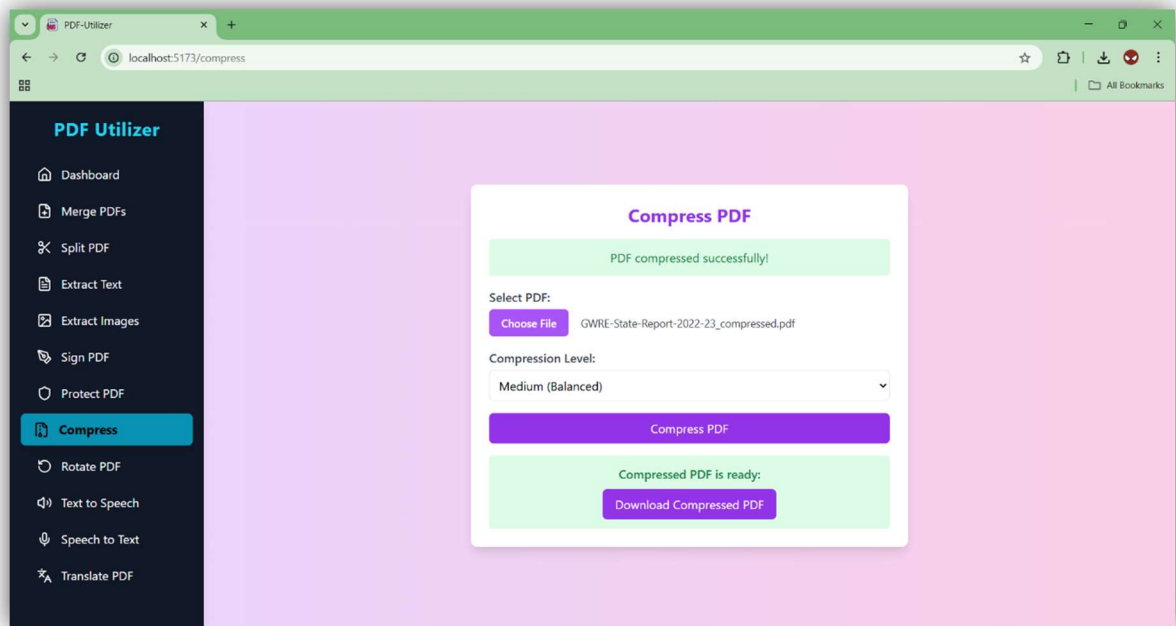
This is the Dark Mode Dashboard of PDF-Utilizer, offering a sleek, user-friendly interface optimized for low-light environments while providing access to all PDF tools in a visually appealing layout.

5.



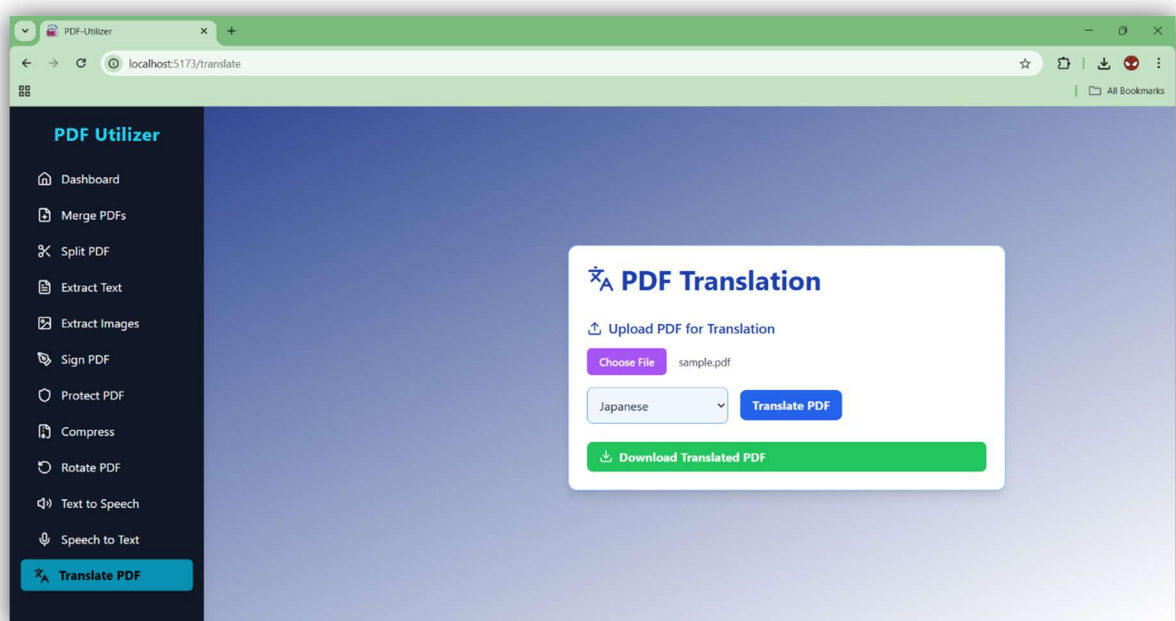
This is the Text-to-Speech (TTS) page of PDF-Utilizer, where users can upload PDF files and convert the text content into natural-sounding audio, making documents more accessible and convenient to consume.

6.



This is the Compress PDF page of PDF-Utilizer, allowing users to reduce the file size of large PDF documents without compromising quality, making them easier to share and store.

7.



This is the Translate PDF page, where users can translate the text content of their PDF files into different languages using integrated translation services.